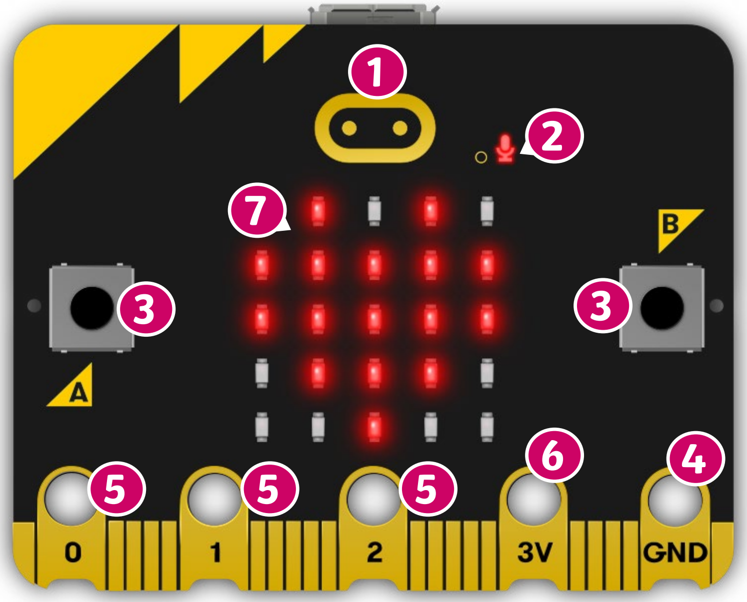
BBC micro:bit

micro:bit the next gen

Front View



1

Touch Sensor

An input device that senses your touch. It can be used as an extra button in your programs.

7

LED Display and Light Sensor

The micro:bit has 25 LEDs arranged in a 5x5 grid. The LEDs can be used for displaying pictures, words and numbers. They can also act as sensors and measure light levels.

6

3V Pin

The 3 volt pin can be used to power accessories connected to the micro:bit.

5

General Purpose Input and Output (GPIO) Pins

The three pins at the edge of the micro:bit labelled 0, 1 and 2. Crocodile clips or strips of foil can be attached to each pin to connect devices such as headphones, buzzers or switches made from cardboard and foil.

2

Microphone

An input device to record sound levels. The microphone LED will light up if the microphone is actively measuring sound.

3

Buttons

The BBC micro:bit has two buttons, A and B on the front. The buttons are input devices and can be pressed separately or together to trigger code.

4

GND Pin

The GND pin is the ground or Earth pin. It can be used to complete a simple circuit when you connect headphones, LEDs or external switches to the micro:bit.

twinkl

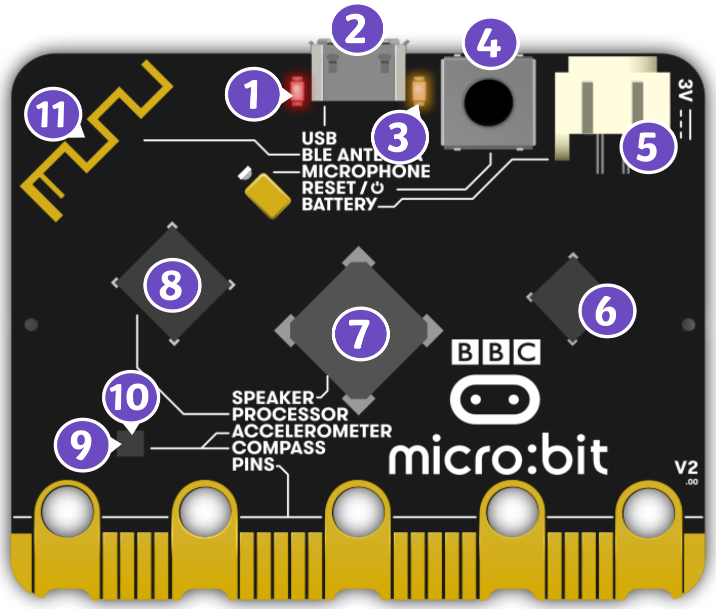
BBC micro:bit

Back View

1 Single LED
Lights up red to show when the micro:bit is being powered by battery or computer.

2 Micro USB Connector
Connect the micro:bit to a desktop computer or laptop with a micro USB cable to power the micro:bit or to transfer programs.

3 Single LED
Flashes yellow when a program is being downloaded on to the micro:bit.



11 Radio Antenna
Allows the micro:bit to send and receive messages wirelessly.

4 Reset Button
Switch the micro:bit on and off using this button.

10 Compass
Detects the direction in which it is facing and measure the strength of magnetic fields.

5 Battery Socket
Connect a battery pack to power the micro:bit instead of a computer. This is handy if you want to take the micro:bit outside.

9 Accelerometer
A motion sensor that detects when the micro:bit is tilted left to right, backwards and forwards and up and down.

8 Processor
The micro:bit contains a microprocessor. The processor receives inputs, runs programs and gives outputs.

The processor also has a temperature sensor built in to it to measure the air temperature.

7 Speaker
An output device to create sounds that can be played through headphones or speakers.

6 USB Interface Chip
The interface chip allows you to connect a desktop or laptop computer to a micro:bit so you can transfer programs to it and power it.